

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I S.Venkateswarlu/192425340_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Venkateswarlu

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	

8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

Smart Healthcare Diagnosis – Complete Solution

Q.No. 1: Smart Healthcare Diagnosis

Given Use Case

A hospital uses an AI system to identify possible diseases from patient symptoms.

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.
- Patients with fever and cough may have flu.
- Patients with flu and body pain may have viral infection.
- Patients with viral infection require medical attention.

1. Problem Analysis and Knowledge Identification

The objective is to use a knowledge-based AI system to determine whether **Ravi has a viral infection** based on the given symptoms and rules.

Entities

Entity	Description
Ravi	Patient
Flu	Disease
Viral Infection	Disease

Medical Attention Required action

Facts

1. Ravi has fever.
2. Ravi has cough.
3. Ravi has body pain.

Rules

Rule 1: If a patient has fever and cough, then the patient may have flu.

Rule 2: If a patient has flu and body pain, then the patient may have viral infection.

Rule 3: If a patient has viral infection, then the patient requires medical attention.

Main Query

Determine whether:

ViralInfection(Ravi)

is true.

2. Predicates, Variables and Constants

Predicates

Predicate	Meaning
Fever(x)	x has fever
Cough(x)	x has cough
BodyPain(x)	x has body pain
Flu(x)	x has flu
ViralInfection(x)	x has viral infection
MedicalAttention(x)	x requires medical attention

Variable

x represents any patient.

Constant

Ravi is a constant representing the specific patient Ravi.

3. Propositional Logic Representation

Let:

- **F** = Ravi has fever
- **C** = Ravi has cough
- **B** = Ravi has body pain
- **L** = Ravi has flu
- **V** = Ravi has viral infection
- **M** = Ravi requires medical attention

Facts

F

C

B

Rules

Rule 1:

$(F \wedge C) \rightarrow L$

Rule 2:

$$(L \wedge B) \rightarrow V$$

Rule 3:

$$V \rightarrow M$$

Complete Propositional Knowledge Base

$$F \wedge C \wedge B$$

$$(F \wedge C) \rightarrow L$$

$$(L \wedge B) \rightarrow V$$

$$V \rightarrow M$$

4. Logical Connectives Used

Symbol	Meaning	Example
$\wedge$	AND	$F \wedge C$
$\vee$	OR	$F \vee C$
$\neg$	NOT	$\neg V$
$\rightarrow$	Implication	$F \rightarrow L$
$\leftrightarrow$	Equivalence	$F \leftrightarrow C$

The main rules use **AND** and **implication**.

For example:

$$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$$

means:

If x has fever AND x has cough, then x has flu.

5. First-Order Logic Representation

First-Order Logic allows us to represent the rules for **any patient**, rather than only Ravi.

Fact 1

$$\text{Fever}(\text{Ravi})$$

Fact 2

$$\text{Cough}(\text{Ravi})$$

Fact 3

$$\text{BodyPain}(\text{Ravi})$$

Rule 1

For every patient:

$\forall x [(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]$

Meaning:

For every person x, if x has fever and cough, then x has flu.

Rule 2

$\forall x [(Flu(x) \wedge BodyPain(x)) \rightarrow ViralInfection(x)]$

Meaning:

For every person x, if x has flu and body pain, then x has a viral infection.

Rule 3

$\forall x [ViralInfection(x) \rightarrow MedicalAttention(x)]$

Meaning:

For every person x, if x has a viral infection, then x requires medical attention.

6. Quantifiers

The universal quantifier $\forall$ means **for all**.

For example:

$\forall x [(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]$

Here, x represents every possible patient.

An existential statement could be written as:

$\exists x ViralInfection(x)$

Meaning:

There exists at least one person who has a viral infection.

After reasoning, we can establish that Ravi is such a person.

7. Unification and Substitution

Unification is the process of making two logical expressions identical by finding suitable substitutions.

Consider the rule:

$Fever(x) \wedge Cough(x) \rightarrow Flu(x)$

and Ravi's facts:

$Fever(Ravi)$

$Cough(Ravi)$

We unify:

$Fever(x)$

with

Fever(Ravi)

Therefore:

$x = \text{Ravi}$

Similarly:

Cough(x)

unifies with:

Cough(Ravi)

Most General Unifier (MGU)

$\{x/\text{Ravi}\}$

After substitution:

$\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi}) \rightarrow \text{Flu}(\text{Ravi})$

Therefore:

$\text{Flu}(\text{Ravi})$

can be derived.

8. Second Unification

Consider Rule 2:

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

Known facts:

$\text{Flu}(\text{Ravi})$

$\text{BodyPain}(\text{Ravi})$

Unification gives:

$x = \text{Ravi}$

Therefore:

$\text{Flu}(\text{Ravi}) \wedge \text{BodyPain}(\text{Ravi})$

$\rightarrow \text{ViralInfection}(\text{Ravi})$

Hence:

$\text{ViralInfection}(\text{Ravi})$

9. Complete Inference Sequence

The AI system uses the following reasoning sequence.

Step 1 – Initial Facts

Fever(Ravi)

Cough(Ravi)

BodyPain(Ravi)

Step 2 – Apply Rule 1

Rule:

$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

Substitution:

$x = \text{Ravi}$

Therefore:

$\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi})$

$\rightarrow \text{Flu}(\text{Ravi})$

So we derive:

$\text{Flu}(\text{Ravi})$

Step 3 – Apply Rule 2

Rule:

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

Substitution:

$x = \text{Ravi}$

Therefore:

$\text{Flu}(\text{Ravi}) \wedge \text{BodyPain}(\text{Ravi})$

$\rightarrow \text{ViralInfection}(\text{Ravi})$

So we derive:

$\text{ViralInfection}(\text{Ravi})$

Step 4 – Apply Rule 3

Rule:

$\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$

Substitution:

$x = \text{Ravi}$

Therefore:

$\text{MedicalAttention}(\text{Ravi})$

10. Resolution Proof

We now prove:

ViralInfection(Ravi)

using resolution.

Step 1 – Convert the rules into CNF

Rule 1:

$(\text{Fever}(x) \wedge \text{Cough}(x)) \rightarrow \text{Flu}(x)$

Using:

$A \rightarrow B \equiv \neg A \vee B$

we get:

$\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

Clause C1

$\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

Rule 2:

$(\text{Flu}(x) \wedge \text{BodyPain}(x)) \rightarrow \text{ViralInfection}(x)$

Clause C2

$\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$

Rule 3:

$\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$

Clause C3

$\neg \text{ViralInfection}(x) \vee \text{MedicalAttention}(x)$

Step 2 – Facts as Clauses

C4

Fever(Ravi)

C5

Cough(Ravi)

C6

BodyPain(Ravi)

11. Resolution Derivation

Derive Flu(Ravi)

Take:

C1: $\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

Substitute:

$x = \text{Ravi}$

We get:

$\neg \text{Fever}(\text{Ravi}) \vee \neg \text{Cough}(\text{Ravi}) \vee \text{Flu}(\text{Ravi})$

Resolve with:

$\text{Fever}(\text{Ravi})$

Result:

$\neg \text{Cough}(\text{Ravi}) \vee \text{Flu}(\text{Ravi})$

Resolve with:

$\text{Cough}(\text{Ravi})$

Result:

$\text{Flu}(\text{Ravi})$

Therefore:

$\text{Flu}(\text{Ravi})$

is proven.

12. Prove $\text{ViralInfection}(\text{Ravi})$

Take:

C2:

$\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$

Substitute:

$x = \text{Ravi}$

Therefore:

$\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$

Resolve with:

$\text{Flu}(\text{Ravi})$

Result:

$\neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$

Resolve with:

$\text{BodyPain}(\text{Ravi})$

Result:

$\text{ViralInfection}(\text{Ravi})$

Therefore:

✓ $\text{ViralInfection}(\text{Ravi})$

has been logically established.

13. Resolution by Refutation

To formally prove the query, assume its opposite:

$\neg \text{ViralInfection}(\text{Ravi})$

We already derived:

$\text{ViralInfection}(\text{Ravi})$

Now resolve:

$\text{ViralInfection}(\text{Ravi})$

with:

$\neg \text{ViralInfection}(\text{Ravi})$

The result is the empty clause:

□

The empty clause represents a contradiction.

Therefore, the assumption:

$\neg \text{ViralInfection}(\text{Ravi})$

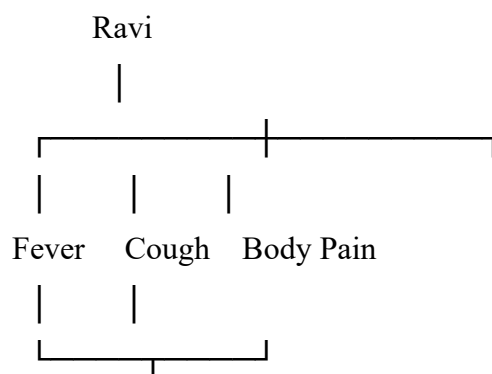
is false.

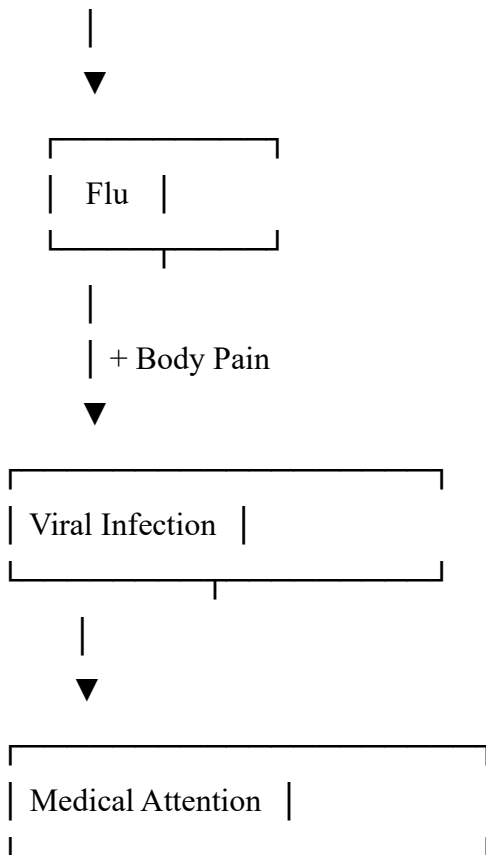
Hence:

$\text{ViralInfection}(\text{Ravi})$

is true according to the given knowledge base.

14. Mapping / Logic Diagram





Compact Mapping

Fever(Ravi) + Cough(Ravi)



Flu(Ravi)



Flu(Ravi) + BodyPain(Ravi)



ViralInfection(Ravi)



MedicalAttention(Ravi)

15. Complete Knowledge Base

The complete FOL knowledge base is:

Facts

Fever(Ravi)

Cough(Ravi)

BodyPain(Ravi)

Rules

$\forall x [(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]$

$\forall x [(Flu(x) \wedge BodyPain(x)) \rightarrow ViralInfection(x)]$

$\forall x [ViralInfection(x) \rightarrow MedicalAttention(x)]$

Query

$ViralInfection(Ravi)$

Inference

$Fever(Ravi)$

+

$Cough(Ravi)$

↓

$Flu(Ravi)$

+

$BodyPain(Ravi)$

↓

$ViralInfection(Ravi)$

↓

$MedicalAttention(Ravi)$

16. Final Conclusion

From the given facts and rules, Ravi has **fever, cough, and body pain**.

Using Rule 1:

$Fever(Ravi) \wedge Cough(Ravi)$

$\rightarrow Flu(Ravi)$

Using Rule 2:

$Flu(Ravi) \wedge BodyPain(Ravi)$

$\rightarrow ViralInfection(Ravi)$

Using Rule 3:

$ViralInfection(Ravi)$

$\rightarrow MedicalAttention(Ravi)$

Therefore, the final logical conclusion is:

$\text{ViralInfection(Ravi)} = \text{TRUE}$
$\text{MedicalAttention(Ravi)} = \text{TRUE}$

Final Answer: According to the given AI knowledge base, **Ravi can be inferred to have a viral infection and therefore requires medical attention.**